Collaborative Online International Learning Project

University of Applied Sciences Mittelhessen

CESI École d'Ingénieurs

Automatic Solar Positioning System and Virtual Power Plant

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1. The CESI – THM COIL Project

COIL means Collaborative Online International Learning. Therefore in the CESI-THM COIL project we practice international collaboration between student groups from the French engineering school CESI and the German University of Applied Sciences TH Mittelhessen.

Collaboration within and between the teams will mainly be online. This means that various software tools will be introduced and used to exchange information and communicate within the project. This is just like in companies and other large organisations today. So, the students learn to use these tools in an effective way. However, they will also learn about the difficulties of working in teams especially, when these teams cannot get in personal contact. Supervisors will support in situations where the teams cannot solve difficulties on their own.

The topic of the project to be handled by the teams is “Photovoltaic energy production and the Internet of Things”. Within this project, various aspects will be investigated:

- Fundamentals and measurement of the characteristics of solar panels
- Different ways of mounting solar panels (static and dynamic)
- Remote control of dynamic solar panel mounting via the Internet of Things
- Combining many photovoltaic generators at different locations to one large photovoltaic power plant

In this **guideline**, the fundamentals of the project as well as the different project tasks are described.

Finally, the student teams will have to summarize their project in a **project report** which answers as many questions from these guidelines as possible and demonstrates the setups and prototypes that have been built.

2. Fundamentals

2.1. Photovoltaic Power Generation

2.1.1. Introduction

Photovoltaic solar energy generation depends on the orientation and inclination of the solar array. Solar energy is produced during the period when the sun's rays are shining directly or indirectly on the panel. It is then said to be intermittent, because electricity is only produced during the day. So how can you make the most of solar energy?

The simplest way to achieve optimum solar energy production is to optimize the orientation and inclination of the photovoltaic panel with respect to the sun.

2.1.2. Path of the Sun – Orientation and Inclination

To better understand solar production, let's look at a few principles of solar radiation. To begin with, let's talk about the sun's path, which is the path the sun takes from a reference point. As we already know, the Earth revolves around the sun, on an elliptical trajectory, but if we position ourselves on a reference point, we can think that it's the sun that revolves around the Earth.

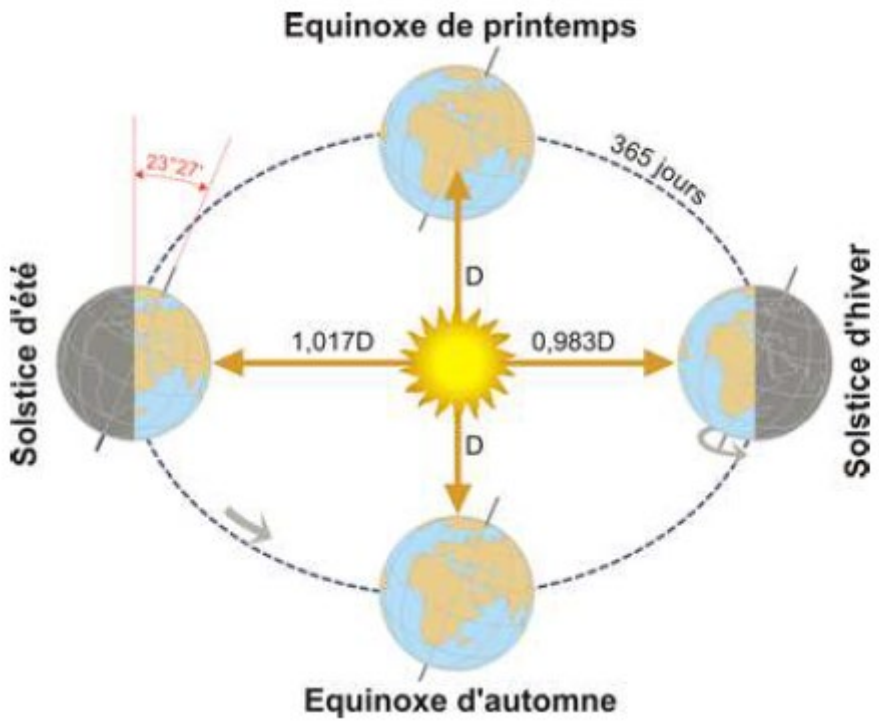


Figure 1: The sun*s apparent path

The Earth rotates on itself in 24 hours at an angle of 23°.27' to the perpendicular to the plane of the ellipse. This inclination is constant throughout its course around the sun and is responsible for seasonal variations. Thus, during our winter months, in the northern hemisphere, the duration of insolation is relatively short, and the sun will not

rise very high in the sky, while summer reigns over the southern hemisphere. During our summer months, the situation is reversed, with the northern hemisphere facing the Sun. Days are then longer than nights in the northern hemisphere, and incident radiation approaches the vertical.

To fully understand and use the sun's influence in the choice and treatment of a site, you need to know the sun's position in the sky at all times. This information is essential for calculating solar gain, choosing the exposure of a building, installing active solar systems (thermal or photovoltaic), designing adjacent exterior areas, natural lighting for interior rooms, positioning windows, solar protection and vegetation, and so on.

At a given moment, the Sun's height and azimuth determine its position in the sky.

- The height α of the Sun is the angle between the direction of the Sun and the horizontal plane. It is counted from 0° to 90° from the horizon towards the celestial vault.
- The Sun's azimuth γ is the angle created between the vertical plane passing through both the Sun and the location under consideration, and the N-S vertical plane. This angle is 0° to the south, and is conventionally positive towards the west and negative towards the east.

The SunEarthTools or PVGIS websites let you visualize the sun's path anywhere in the world, at any time of the year.

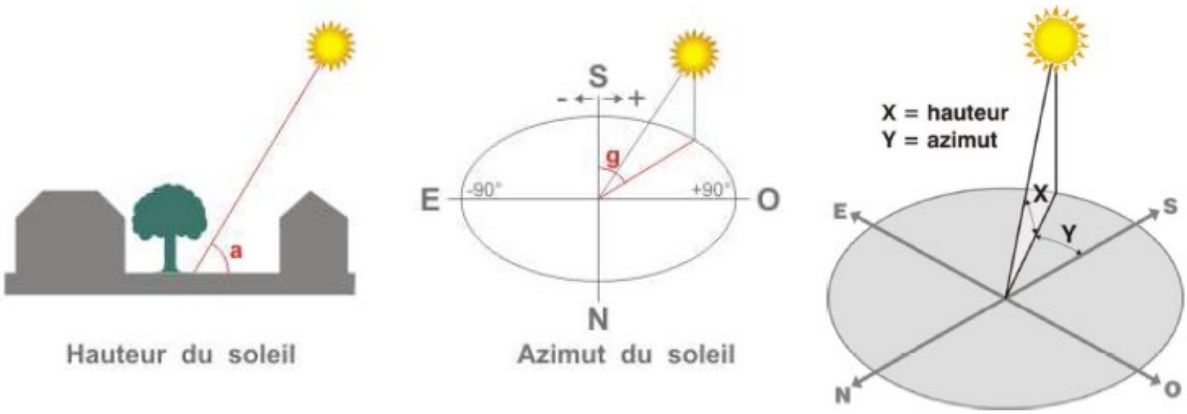


Figure 2: Representation of the sun's height and azimuth.

For greater clarity, the solar path is usually represented by a diagram in rectangular coordinates.

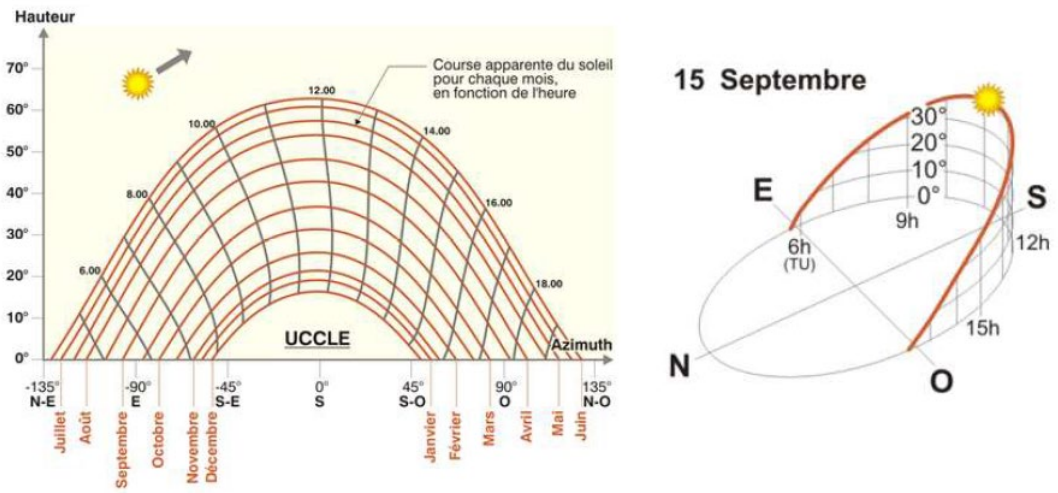


Figure 3: Cylindrical solar diagram for Uccle in Universal Time and example of Azimuth and Height for September 15

2.1.3. Position of Solar Panels

The right positioning of solar panels increases the overall yield of a PV (PhotoVoltaic) system by increasing the solar energy received. Improving capture efficiency by choosing the right orientation for the surface of the modules in the photovoltaic field is therefore an important parameter when setting up a photovoltaic system.

Solar panels should be positioned perpendicular to the sun, facing south when in the northern hemisphere, and north when in the southern hemisphere. They need to be tilted at an "optimal" angle for optimum electricity production.

2.1.4. Incident Solar Irradiance

The angle at which the Sun's rays strike a surface determines the energy density received by that surface. Since solar radiation reaches the Earth in a parallel beam, a surface perpendicular to these rays intercepts the maximum energy density. And if the surface is tilted from this perpendicular position, its irradiance decreases.

Perhaps the best way to represent this phenomenon is to depict the Sun's parallel rays with a handful of pencils held in the hand above a sheet of paper, points down. The marks made by the points represent grains of energy. When the pencils are perpendicular to the sheet, the points are squeezed to the maximum: the density of energy per unit area is greatest. When all these parallel pencils are tilted together, the points spread out and cover increasingly elongated surfaces: energy density decreases as the traces spread out. The angle the Sun's rays make with the surface normal (angle of incidence) will determine the percentage of direct light intercepted by the surface.

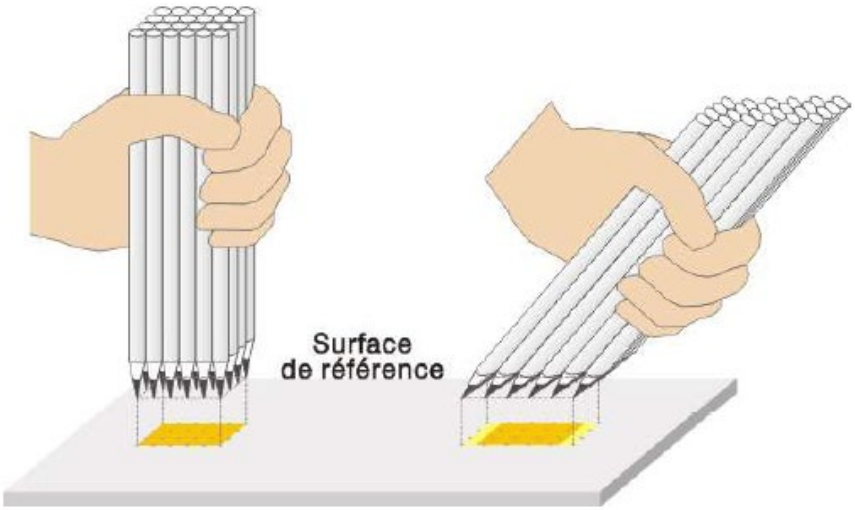


Figure 4: Representation of incident solar irradiance

2.1.5. Air Mass

Air mass", m, is the mass of atmosphere through which direct radiation passes to reach the ground (under clear skies), compared with a vertical crossing at sea level.

- AM0 corresponds to irradiation outside the atmosphere.
- AM1.5 D represents direct irradiation.
- AM1.5G (global) takes direct and diffuse radiation into account

Example:

- Sun at zenith (at sea level h=900): AM1.
- Sun to 300: AM2.
- Sun at 480: AM1.5.

And by convention, AM0 refers to solar radiation outside the atmosphere.

2.1.6. Definitions of Photovoltaic Cell Parameters

- **Peak power**, **Pc**, is the electrical power the module can deliver under standard conditions: temperature 25°C, illuminance 1000 W/m2 and air mass of 1.5 (see Appendix A).
- **The short-circuit current Isc** is the current obtained when zero voltage is applied across the solar cell, i.e. when the load resistance is zero. It represents the intersection of the characteristic with the "Isc" current axis.
- **The open circuit voltage Voc** is the voltage that appears across the cell when the load is infinitely large. It represents the intersection with the voltage axis in the characteristic shown in Figure 5.

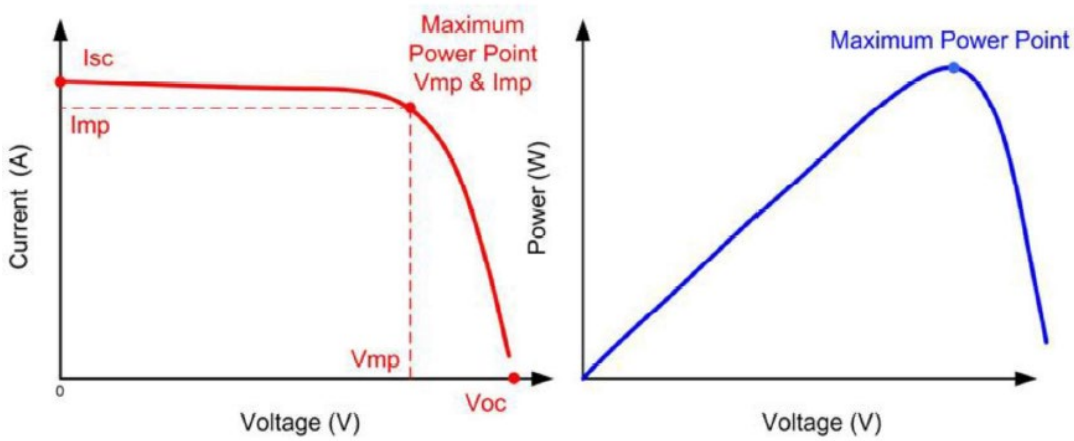


Figure 5: Electrical characteristics of a solar panel

- **Maximum power P_{mp}**: when the solar cell is connected to a load, it behaves like an electrical power generator. As shown in Figure 5, the point where the load line intersects the 1 = f(V) characteristic corresponds to maximum power, which is the product of maximum voltage and maximum current. Pmp = Vmp x Imp
- **Efficiency of the solar module η:** The efficiency of the solar module is the maximum electrical power generated by the solar module deviced by the related solar power encountered by the module. It is given in percent.

  $$\eta = \frac{P_{mp}}{P_{sol}} [\%]$$

  The solar power can be measured with a solarimeter and must be referred to the area of the solar module.
  Typical efficiencies of solar modules today are between 18% and 24%.
- **The Form Factor FF** is the optimal operating point of the photovoltaic cell, i.e. the point at which maximum power is delivered to the load. This power increases as the ratio, known as the form factor, approaches unity, such as :

$$FF = \frac{V_{max} \cdot I_{max}}{V_{co} \cdot I_{cc}}$$

2.1.7. Solar Module Orientation and Tilt

The position of PV modules in relation to the sun directly influences their energy production. It's very important to position them correctly to use them to their full potential.

Orientation refers to the cardinal point towards which the active face of the panel faces (south, north, south-west...).

As for **inclination**, it indicates the angle the panel makes with the horizontal plane and is therefore counted in degrees.

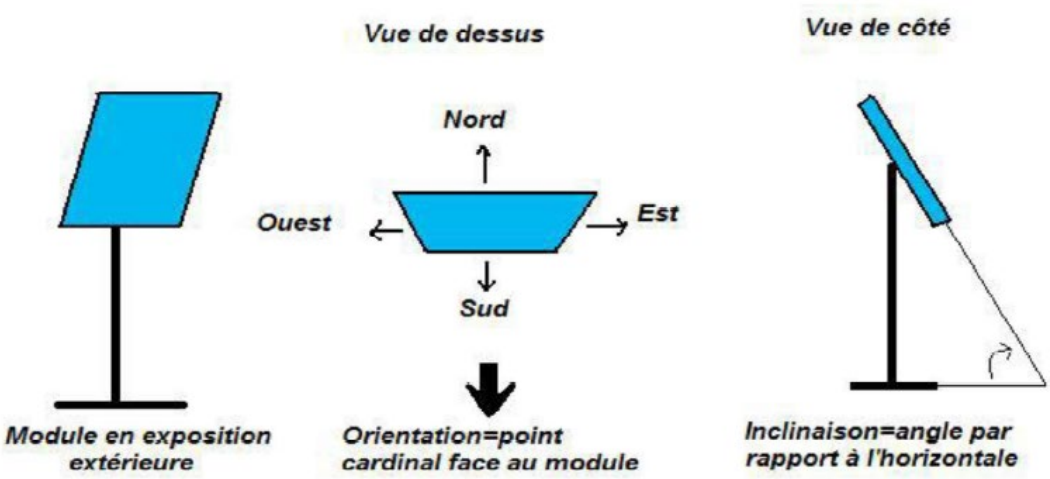


Figure 6: Defining the orientation and inclination of a solar panel.

2.1.8. Energy Production with Solar Panels

As we have just seen, the sun moves continuously throughout the day (and year). Today, however, almost all photovoltaic energy production systems are based on fixed solar panels, which results in a significant loss of energy that could be available and greatly reduces the production efficiency of these systems. However, there are several different photovoltaic power generation systems available today:

- Fixed Panels
- Manual Tilt and fixed Orientation
- Dynamic Orientation or Tilt
- Dynamic Tilt and Turn

2.1.9. Fixed Panel

In a fixed installation (on a roof, on stilts...) yield is optimized by exposing the panels due south, and the energy supplied by the PV modules is at its maximum only when the sun is at its zenith. Of course, the inclination is adapted to the geographical position on earth, to take into account the inclination, azimuth and height of the sun...

2.1.10. Manual Tilt, Fixed Orientation

For a system that can be manually oriented with 1 degree of freedom, you can, for example, change the tilt every month, but for ease of use you can also choose one tilt for summer, another for winter, to avoid repetitive handling of the photovoltaic panel tilting systems. This type of structure is generally used for ground-mounted applications or on flat roofs, to best adapt to the position of the sun. For these variable- tilt systems, the panels are tilted at an angle to the horizontal to maximize monthly electricity production.

2.1.11. Dynamic Orientation or Tilt

For an automated 1-degree-of-freedom orientation system, we can of course make systems that adapt the inclination in real time to the position of the sun, or modify the orientation of the panel in real time. This kind of system optimizes energy production. However, servo-control of this type of system requires motorized structures that consume electrical power and require electronic control.

2.1.12. Dynamic Tilt and Turn

Photovoltaic modules placed on trackers that are dynamic in terms of inclination and orientation therefore produce significantly more electricity than previous installations. This setup involves placing photovoltaic panels on structures known as solar tracking systems or solar trackers. As a result, the PV cells are systematically perpendicular to the incident radiation, allowing the system to operate in optimal conditions. Although they are not a necessary part of a photovoltaic system, their use can dramatically improve the power output result by tracking the sun throughout the day.

Figure 7: EDF Smartflower (Solar Tracker)

2.1.13. Production Comparison

A comparison is made between the fixed mounting and the solar tracker for the same type of 1.1 kWp panels (produced). On a fully sunny day, a properly oriented 1.1 kWp

Figure 8: Hourly production comparison between a fixed and a mobile 2-axis system.

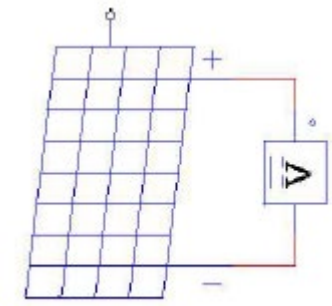
system produces 5.5 kWh of energy, while the same system with tracker, under the same sunshine conditions, produces 11 kWh of energy.

2.1.14. Characterization of a Solar Panel

The U-I characteristic of the PV module must be measured at constant irradiance (if solar) or illuminance (if lamp). In the event of minimal variation, you can take an average, for example. In the event of significant change (variable weather, clouds causing shading, etc.), the results will not be usable (this will be reflected in the plotted characteristic).

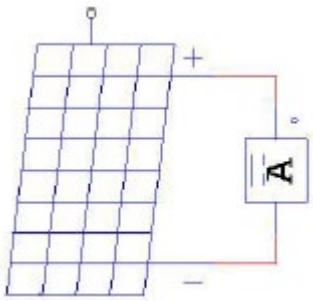
Open Circuit Voltage Measurement

Open-circuit voltage V_{oc} of a solar panel can be measured by connecting a voltmeter (DC) directly to the PV module terminals.



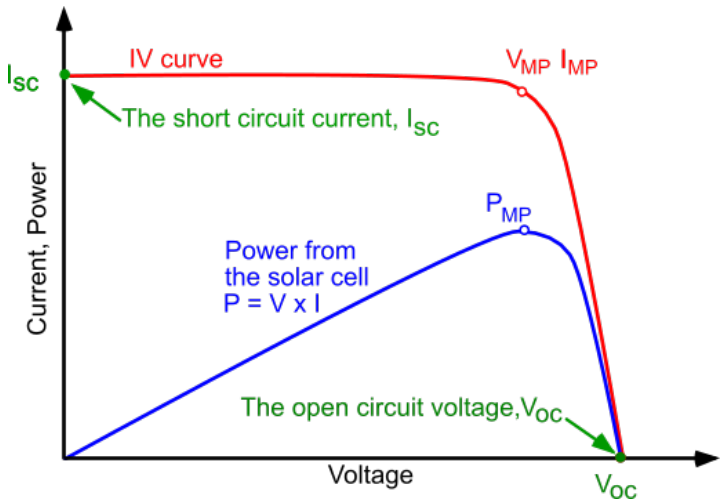
Short Circuit Current Measurement

Short-circuit current I_{sc} of a solar panel can be measured by feeding the PV module directly into an amperemeter (direct current).



Measurements to plot the voltage-current Characteristic of the Module

If you plot the current over the voltage delivered by the panel, you get the characteristic IV curve of a solar panel. If you plot the power generated by the solar panel, you get the power characteristic curve of the solar panel.

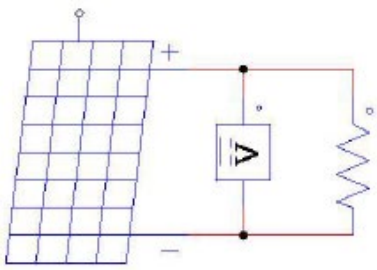


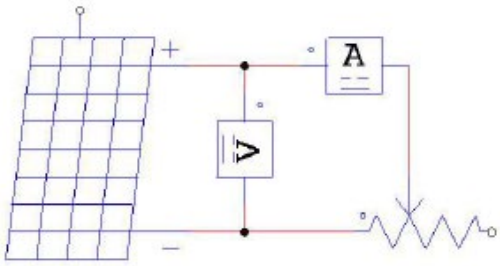
The IV characteristic can be plotted by feeding the PV module through a load resistor R and determining the value of the current I delivered by the PV module and the value of the voltage across its terminals. The end points of the curve are the Isc and the Voc. There are two ways of doing this:

- Using **two multimeters**, one connected as an amperemeter (direct current) in series in the circuit, and the other connected as a voltmeter (direct current) at the terminals of the PV module. (If desired, the resistance value can be determined via R = U/I).
- Using a **single multimeter**:
  - Measure resistance value R beforehand, then
  - Measurement of the voltage U across the PV module via the voltmeter (direct current), in which case / = U/R, or
  - Measure the current / in the circuit using the ammeter (direct current), in which case
    U = R*I

Then vary the value of the resistor R:

- By using a potentiometer (variable resistance). This solution may be more suitable if you have two multimeters for measurements (although you can still systematically measure resistance beforehand, as indicated above).
- Alternatively, by using a conventional set of resistors, possibly combining them in series and/or parallel to obtain resistance values not available in the original set. Even if resistors are standardized, it's always preferable to measure their "true" value.





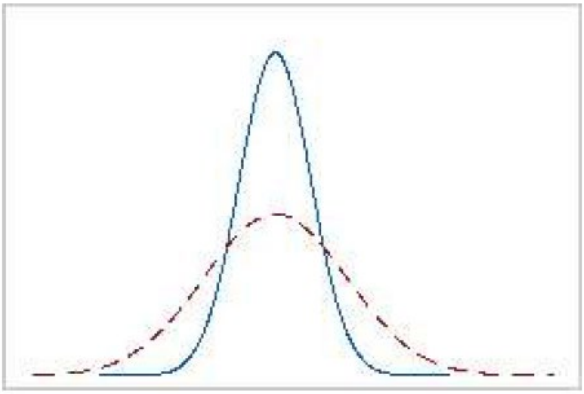
2.2. Repeatability and Reproducibility

Repeatability and reproducibility are the two components of precision in a measuring system. To assess repeatability and reproducibility, let's take the example of a gauge study.

Repeatability

Repeatability is the variation due to the measuring instrument. It is the variation observed when the same operator measures the same part many times, using the same instrumentation, under the same conditions.

To estimate the repeatability of a measurement device A, operator 1 measures a single part with the A device 20 times. To estimate the repeatability of a measurement device B, Operator 1 measures a single part with the B device 20 times. Instrumentation A shows less variation; it is therefore more repeatable than instrumentation B.

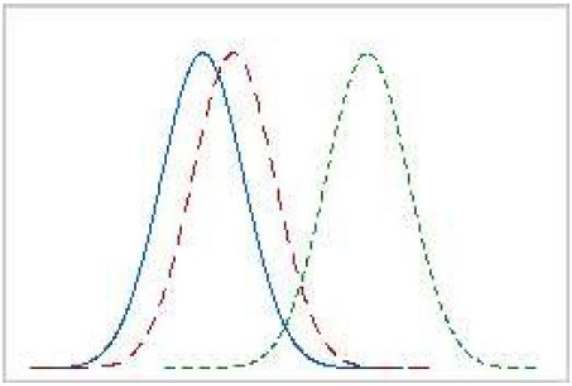


The solid line is made up of A device measurements. The dotted line corresponds to B device measurements.

Reproducibility

Reproducibility is the variation due to the measurement system. It is the variation observed when different operators measure the same part many times, using the same instrumentation, under the same conditions.

Example: operators 1, 2 and 3 measure the same part 20 times using the same instrumentation.



The three lines are the measurements of operator 1, 2 and 3. The variation in average measurements between operator 1 and operator 2 is much smaller than the variation between operators 1 and 3. As a result, the gauge's reproducibility is too low.

2.3. Internet of Things

2.3.1. Introduction

The Internet of Things (IoT) refers to a network of interconnected physical devices that communicate and exchange data over the internet. These devices, often equipped with sensors, actuators, and communication technologies, collect, transmit, and process data to enable automation, efficiency, and decision-making across various domains. IoT integrates both the digital and physical worlds, allowing real-time monitoring, remote control, and smart interactions between objects and systems.

The Internet of Things is a transformative technology that bridges the physical and digital worlds through connected devices and intelligent data processing. By integrating IoT components such as sensors, networks, cloud computing, and AI, businesses and individuals can enhance automation, efficiency, and decision-making across various sectors. However, challenges such as data security, privacy concerns, and interoperability must be addressed to ensure sustainable and secure IoT adoption in the future.

Key Components

The IoT ecosystem consists of several essential components that facilitate its operation:

- Devices and Sensors: IoT devices include smart sensors, wearables, home appliances, industrial machines, and embedded systems that collect and transmit data.

- Connectivity: Various communication protocols enable IoT devices to connect to networks. These include Wi-Fi, Bluetooth, Zigbee, LoRaWAN, 5G, MQTT, CoAP, XMPP and traditional cellular networks.
- Edge Computing and Gateways: To process data locally and reduce latency, edge computing devices and gateways filter, analyze, and transmit relevant data to cloud or on-premise servers.
- Cloud Computing and Data Storage: Large volumes of IoT data are stored and processed in cloud platforms such as AWS IoT, Microsoft Azure IoT, or Google Cloud IoT to facilitate analysis and decision-making.
- Data Analytics and Artificial Intelligence (AI): Advanced analytics and AI algorithms help derive insights from IoT data, enabling predictive maintenance, anomaly detection, and automation.
- Security and Privacy: IoT security includes encryption, authentication mechanisms, and secure communication protocols to protect data and prevent cyber threats.
- User Interfaces (UI) and Applications: Web and mobile applications provide users with dashboards, alerts, and control interfaces to interact with IoT systems effectively.

Application Areas

IoT technology has revolutionized multiple industries by enhancing efficiency, reducing costs, and improving user experiences. Key application areas include:

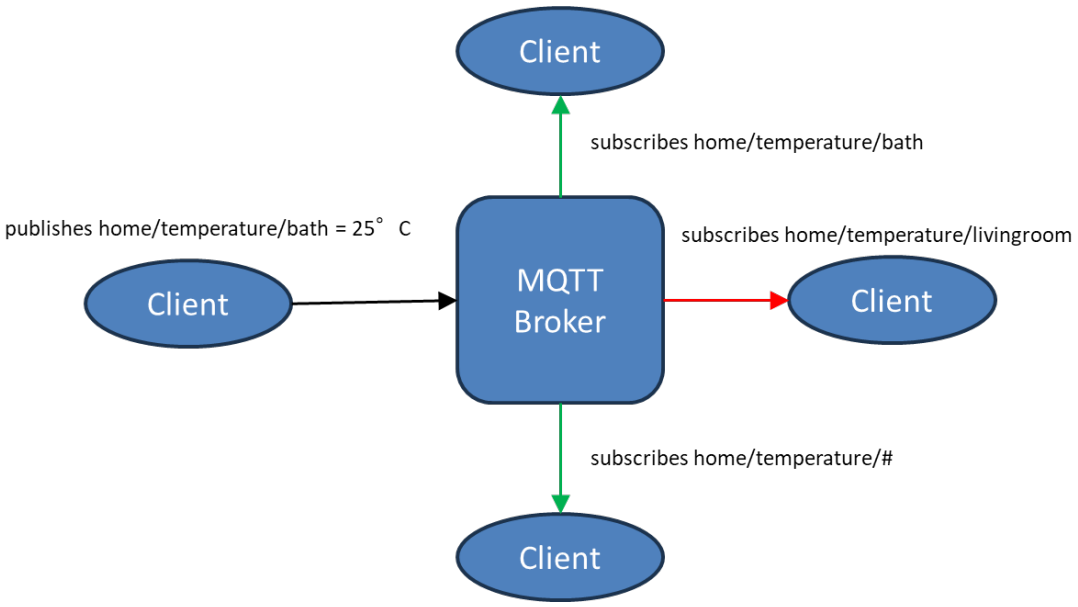
- Smart Homes and Buildings: IoT enables smart lighting, heating, security systems, and voice-controlled assistants (e.g., Amazon Alexa, Google Nest) for automation and energy efficiency.
- Healthcare and Wearable Technology: IoT devices such as smartwatches, fitness trackers, and remote patient monitoring systems help in health tracking and medical diagnostics.
- Industrial IoT (IIoT) and Manufacturing: Predictive maintenance, automation, and real-time monitoring improve productivity and reduce downtime in industrial settings.
- Smart Cities: IoT solutions optimize traffic management, waste management, smart parking, and public safety through connected sensors and analytics.
- Agriculture and Environmental Monitoring: IoT-based smart farming involves precision irrigation, soil monitoring, and climate control for efficient resource use.
- Automotive and Transportation: Connected vehicles, fleet tracking, and autonomous driving leverage IoT for enhanced navigation, safety, and logistics.
- Retail and Supply Chain: IoT applications in retail include smart shelves, automated inventory management, and personalized customer experiences.

2.3.2. MQTT Communication on the Internet of Things

MQTT (Message Queuing Telemetry Transport) is a lightweight messaging protocol specifically designed for constrained devices and low-bandwidth, high-latency, or unreliable networks. It follows a **publish-subscribe communication model**, making it an efficient choice for IoT applications.

Key Features of MQTT:

- Publish-Subscribe Model: Unlike traditional client-server models, MQTT uses a broker to manage messages between clients, enabling efficient and scalable communication.
- Lightweight Protocol: MQTT minimizes overhead with small message headers, making it ideal for low-power devices.
- Physical Layer: MQTT can be used on Ethernet (wired) as well as WiFi (wireless) networks.
- Client-Broker Pattern: Clients publish information under a certain topic (see below) or listen to published information under a certain topic. The Broker distributes the information within the Network.



- Quality of Service (QoS) Levels: Ensures reliable message delivery with three levels:
  - QoS 0: At most once (fire-and-forget)
  - QoS 1: At least once (acknowledged delivery)
  - QoS 2: Exactly once (guaranteed delivery)
- Persistent Sessions: Clients can resume communication without losing messages if they disconnect temporarily.

2.3.3. MQTT Topics, Subtopics, and Wildcards

MQTT organizes messages into topics, allowing clients to publish and subscribe efficiently. Topics are a hierarchical structure (e.g., home/temperature/livingroom) that defines message routing. Within that structure, **wildcards** can substitute one or more levels:

- + (Single-level wildcard): Matches one level in the hierarchy (e.g., home/temperature/+ matches home/temperature/livingroom).
- # (Multi-level wildcard): Matches multiple levels (e.g., home/# captures all topics under home).

Please notice that topics in MQTT are always lowercase.

In this project we use the following topic structure for MQTT messages:

Topic	Description
cesithmcoil2025/	main topic
cesithmcoil2025/test	for general test purposes, subtopics can be defined by users, testclient
cesithmcoil2025/<teamname>/<city>/#	for messages within a certain team, the city relates to the location where the message is from.
cesithmcoil2025/<teamname>/<city>/test/#	for test messages within the team
cesithmcoil2025/<teamname>/<city>/h-angle	Target position of the panel defined by horizontal angle
cesithmcoil2025/<teamname>/<city>/v-angle	Target position of the panel defined by vertical angle
cesithmcoil2025/<teamname>/<city>/panelcurrent	Actual current generated by the panel [mA]
cesithmcoil2025/<teamname>/<city>/panelvoltage	Actual voltage generated by the panel [V]
cesithmcoil2025/<teamname>/<city>/panelpower	Actual power generated by the panel [mW]

2.3.4. MQTT Broker

As was mentioned earlier, MQTT communication always requires an MQTT broker which distributes the messages between the clients. Therefore, the broker is a central component of an MQTT network.

There exist several open MQTT broker on the internet. Some of them are listed beneath:

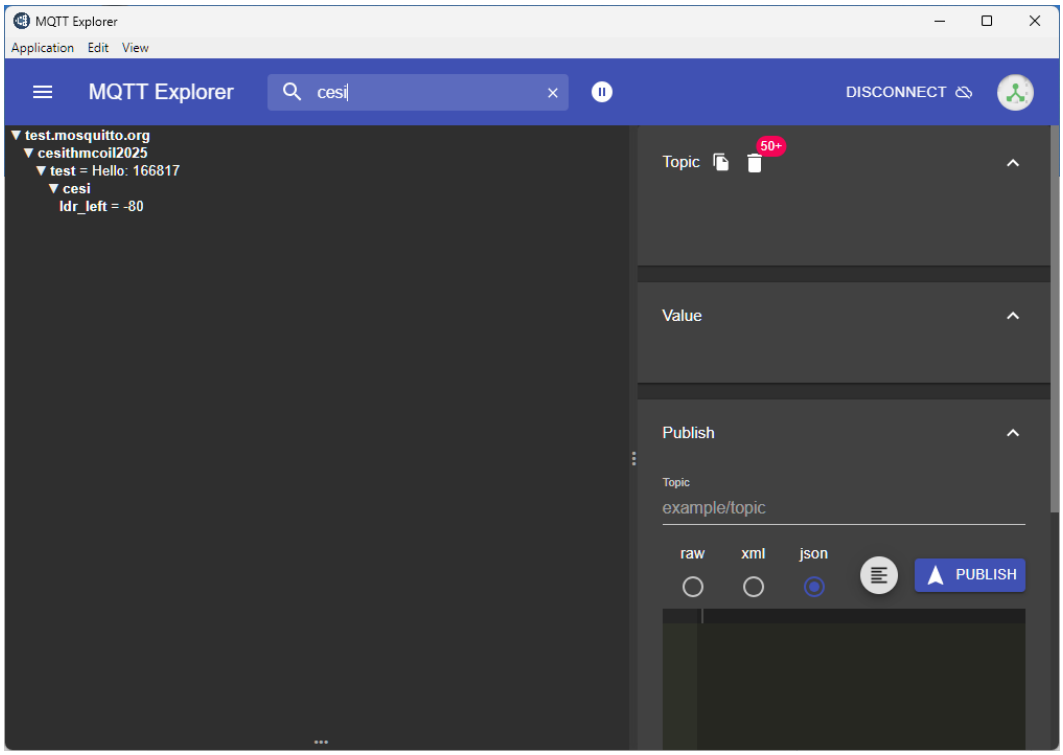
MQTT Broker	IP Adress (URL)	Information
EMQ X MQTT IoT Cloud	https://www.emqx.com/en/mqtt/public-mqtt5-broker	
HIVEMQ	https://www.hivemq.com/public-mqtt-broker/	Download the TLS certificate here
MOSQUITTO	https://test.mosquitto.org/	This MQTT Broker is used in this project on port 1883.
CLOUD MQTT	https://www.cloudmqtt.com/	

Within this project we will use the open **Mosquitto Test Broker**, which can be reached from both universities networks. Its URL is given in the table above. You will have to connect to this broker on port 1883 to exchange unauthenticated and non-encrypted messages. So, the complete URL is: http://test.mosquitto.org:1880

2.3.5. MQTT Tools

A very useful tool if you work with MQTT is the **MQTT Explorer** by Thomas Nordquist. With this tool you can monitor the topics and messages on your MQTT broker, or you can inject messages under a certain topic.

The tool can be downloaded here: https://mqtt-explorer.com/



2.3.6. MQTT Test Client

For this project, we will provide you with a test client, which will send positioning messages via MQTT. You can use them to test the remote control of your solar tracker.

To access the test client, connect your system with the MQTT Test Broker as described above. The positioning messages can then be received under the following topics:

Topic	Meaning
cesithmcoil2025/test/testclient/angle-h	Horizontal angle as an integer number. The value is 90…270 with 90 being direction east, 180 being direction south and 270 being direction west. When 270 degrees have been reached, the value will be set back to 90.

	The value will be increased by 5 degrees every 3 seconds. This topic can be used to test the remote control of the panel direction.
cesithmcoil2025/test/testclient/angle-v	Vertical angle (tilt) as an integer number. The value is 0…90 with 0 being parallel to the earth surface and 90 being perpendicular to the earth surface. When 90 degrees have been reached, the value will be set to 0. The value will be increased by 2 degrees every 3 seconds. This topic can be used to test the remote control of the panel tilt.
cesithmcoil2025/test/testclient/panel-current	Random float value between 0 and 0.4 that simulates the short circuit current I_{SC} of the solar panel in [A]. The value is changed every 2 seconds. The value is not related to other values of the test client. This topic can be used to calculate the power generated by a solar panel.
cesithmcoil2025/test/testclient/panel-voltage	Random float value between 0 and 5 that simulates the open circuit voltage V_{OC} of the solar panel in [V]. The value is changed every 2 seconds. The value is not related to other values of the test client. This topic can be used to calculate the power generated by a solar panel.
cesithmcoil2025/test/testclient/sun-height	Float value that represents the height angle of the sun on 5th of March 2025 at Friedberg (Hesse). The value changes every 5 seconds, simulating the next hour. At 0am the value is -45.08. This topic can be used to calculate the optimum position of a solar panel.
cesithmcoil2025/test/testclient/sun-direction	Float value that represents the direction angle of the sun on 5th of March 2025 at Friedberg (Hesse).

	The value changes every 5 seconds, simulating the next hour. At 0am the value is 347.09. North is represented by the value 0 or 360. This topic can be used to calculate the optimum position of a solar panel.
cesithmcoil2025/test/testclient/ldr-1	Random integer number between 0 and 1000 representing a measurement from a light dependent resistor (LDR). The value is changed every 8 seconds.

Screenshot showing the topics of the Testclient.

2.4. Microcontroller Boards and Arduino IDE

An Introduction to Arduino Microcontroller Boards and their programming will be available as a video in the Moodle course and the library.

2.5. Node Red

An Introduction to Node Red and example files will be available as a video in the Moodle course and the library.

3. Project Tasks

3.1. Teambuilding

During the first meeting we will show you who will be in which team. You will then be asked to prepare a team poster and present it to all the other teams during the next meeting. Your supervisors will support you during the poster preparation.

Your team poster must meet the following requirements:

- The poster must be presented digitally using Zoom as the videoconference platform.
- The poster must introduce each team member as well as the complete team.
- You must be able to upload the team poster into moodle prior to your presentation.

Your team poster must cover the following content:

- An interview with each team member about their person (age, home town, interests and hobbies, studies, favourite place, wishes for the project etc.)
- An overview of specific competences and strengths of your team.
- A team name and a logo representing your team.
- An introduction to your study towns and campuses. Recommendations for the other team members visiting your location. This might include a short video that you produce together using smartphones.
- An overview of how you will communicate within your team (regularly, ad hoc).
- Some team rules and penalties if they are not followed by team members.

You must upload the team poster prior to the second meeting. During the second meeting, you will have 5 minutes time to present your team using the poster.

3.2. Project Planning and Steering

During the second meeting, you will get the task to plan your project and prepare the project planning document. This document should include the following parts:

- A structured list of project tasks derived from the project guideline and the moodle course (Work Breakdown Structure).
- A project milestone plan
- A detailed project schedule (Gantt Type)
- A description of competences within the team and how you will use these competences.
- A description how you will distribute tasks over the two locations (Germany, France).
- An overview of periodic meetings in your team (day, time, cycle, topics, participants)

- An overview of roles in your team. There must be one communicator and one organizer per location. Responsibilites of these persons must be described.
- An agreement on when and how you will escalate issues to the supervisors and who is responsible for escalating.
- An agreement on how you will follow up tasks and report task fulfilment or difficulties. Please review the platform Trello for this.

You will have to upload one project planning document in Moodle. This document has to include all the documentation related to the points above.

3.3. Technical Project on Photovoltaic Energy Production

When we take a quick look at the different photovoltaic energy production systems, it's easy to see that dynamic systems have higher yields, but they're not the ones we're developing in our home or industrial installations. Of course, the different systems have different costs, maintenance requirements and technological challenges, but they all have one thing in common: photovoltaic panels and the production of electrical energy. At present, several technical questions can be asked about these different systems:

- Which parameter has the greatest influence on panel performance: tilt, orientation or a combination of the two?
- Is the power consumption to control a degree of freedom (tilt o orientation) significant compared to the photovoltaic energy production gained by a fixed system?
- What about the two-way mobile tracking system?
- Is it possible to build a solar tracker which is remote controlled over the internet?
- Is it possible to control the position of the solar panel manually and/or automatically over the internet? Which advantages or disadvantages does this have?
- Is it possible to combine various small solar power generators at different locations to virtual large solar power generator? Does this have advantages in terms of uniformity of the total generated power?

At the same time, setting up an international multi-site collaboration to validate experiments and results also raises non-trivial questions:

- How can we ensure that all those involved in the project carry out the experiments correctly?
- Can we analyse and produce conclusions without mastering the repeatability and reproducibility?
- Which difficulties arise in an international virtual team working on a complex project? Which solution approaches did you come up with and how successful were they?
- How can we improve the next international collaboration (equipment, experience, communication, analysis, etc.)?

The aim of this project and the CES/-THM collaboration is to answer these questions and maybe raise new ones.

3.4. Tasks of the technical Project

During this project, you should carry out the following tasks as a team. You start with characterizing the solar module provided by the supervisors.

- Build a 3D printed panel holder with variable tilt (0° to 90° in steps of 5°)
- Measure the power generated by the solar panel using one or two Multimeters and variable load resistors. Generate the characteristic curve of the solar panel at different tilt angles. Measure the solar irradiance using a solarimeter. Find the maximum power point MPP of your solar panel. What is the corresponding current and voltage for your solar panel?
- Find the optimum tilt for your location and a suitable load resistor by analyzing the characteristic curves.
- Fix the tilt and the load resistor at the optimum value and measure the power generated by your solar panel for 30min between 12.30 and 13.00. Note down the power generated every 5 minutes. Measure the solar radiation with a solarimeter.

After you have characterized the solar module, you continue with the solar tracker.

- Build a motorized solar panel positioner (SPP)
- Connect the INA219 measurement board to the solar panel. Use a fixed load derived from the MPP measured earlier.
- Connect the LCD display to your board, if a display has been provided. Otherwise use the serial monitor on your computer.

Then, you build a connected solar tracker using the Internet of Things.

- Connect your SPP to the MQTT broker using the correct topics and control the position of your teammates SPP over the internet using a node red dashboard provided to you. Try to position the solar panel in an optimum way. Which difficulties do you encounter? Does the positioning work?
- upgrade your SPP to a locally controlled automatic solar tracker using the INA219 metering board. Develop an algorithm to quickly and correctly position the solar panel for maximum power generation.
- Measure the current, the voltage and the generated electrical power of your solar tracker and transmit it to the MQTT broker using the correct topics.
- Calculate the overall power of all trackers (virtual power plant with solar trackers). Monitor the total power over a timespan of 30 minutes.

3.5. Tips for setting up and conducting experiments

- It is essential to divide the different experiments among the various national and international collaborators (time constraints).
- It is highly recommended to design a common experimental protocol and to control the experimental conditions as much as possible, to avoid making the experimental results unusable (even if the same experiment is conducted by different collaborators).
- It is strongly advised to repeat quick experiments multiple times within the same team to obtain results with uncertainties.
- Be careful when wiring to properly identify which servomotor has a 180° rotation angle and which one has a 90° angle in relation to your Solar Tracker prototype.
- In the Arduino program, potentiometers are used to adjust the tilt and orientation of the photovoltaic panel. Adding a capacitor slightly stabilizes the servomotors, but it is important to note that the servomotor power is supplied by the Arduino board (+5V). The board is pushed to its limits to power both servomotors, which can cause significant vibrations. Poor connections (wires – breadboard – Arduino) can also lead to system instability. Do not hesitate to stabilize the system with your hands after adjusting the potentiometers.
- It is also possible to manually move the system (disconnected from the Arduino) to take power measurements.
- It is possible to unplug one servomotor to significantly improve the stability of the system.
- The tilt and orientation angles can be measured using a protractor (the measurement method needs to be defined).
- The height (h) of the lamp and the distance (d) from the photovoltaic panel (PV) must be defined to properly determine the tilt and orientation angles.
- It is necessary to eliminate external light sources from the experiment (blackout panel, room in darkness, etc.) if you are measuring with a lamp.
- The power provided by the solar panel is calculated using the measurements obtained from the voltmeter and ammeter. Make sure to connect both measuring tools properly.

- It is possible to produce a PCB for the measurements. You can do so using the software KiCAD and producing the PCB on a milling cutter at THM. The PCB can then be sent to the CESI location by mail.

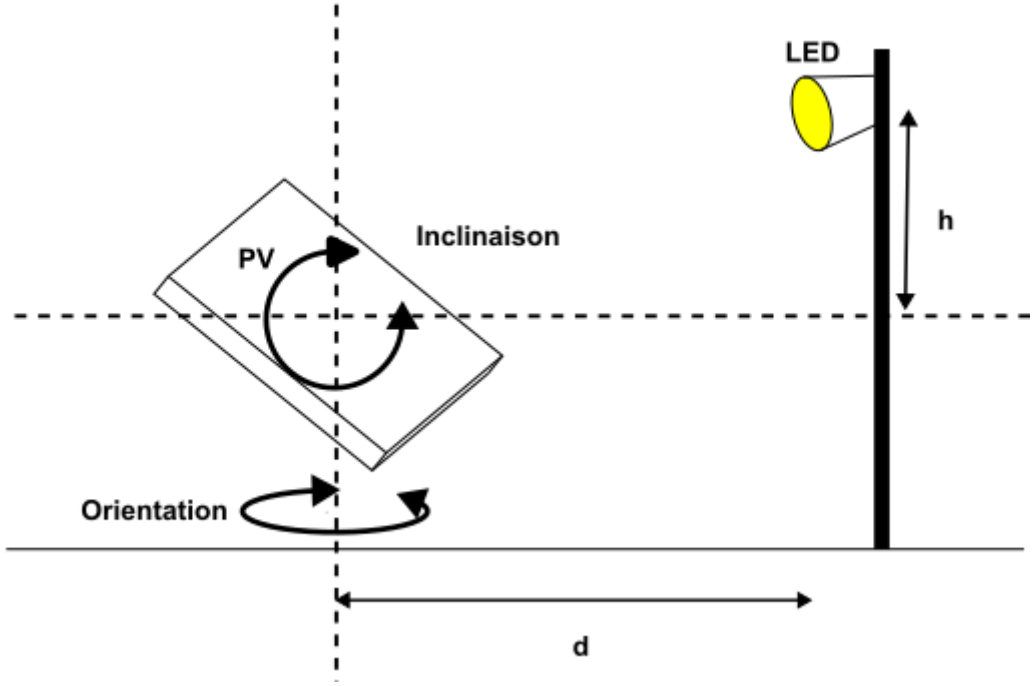


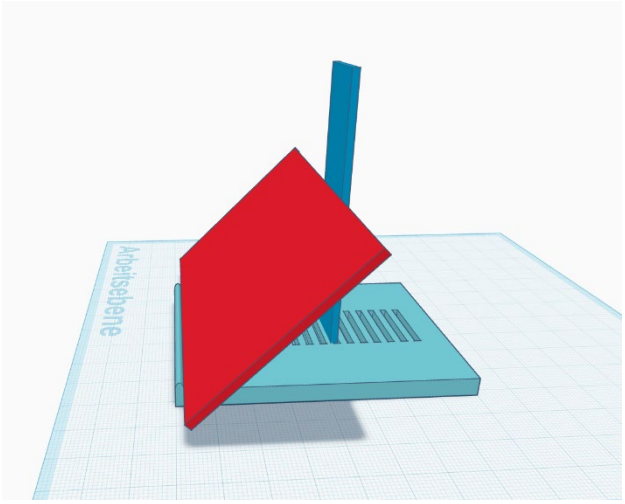
Figure 1 Schematic diagram of the experiment

3.6. Task 1: 3D printed Solar Panel Holder

Construct and build a solar panel holder that allows you to adjust the tilt of your panel between 0° (flat) and 90° (upright standing) in steps of 5 degrees.

Use TinkerCad to construct this panel holder together with a teammate from the other location. Print the panel holder on a 3D printer at your location.

Link: [TinkerCAD]



Example of a panel holder. You are free to construct a different type of holder.

3.7. Task 2: Characterization of the Solar Panel at different tilt angles

Tasks: Direct the solar panel towards a light source (lamp or sun) and measure the characteristic curve IV-curve at different tilt angles (10° variation) using fixed resistors. Document the measurement results using the table below.

Tilt: _____ degrees (vary from 0° to 90°)

Solar irradiance: _____ W/m^2

Direction of the panel: _____ degrees (0° to 360°)

Load Resistor	Current I	Voltage U	Power P
5 Ω			
1000 Ω			

Calculated MPP and corresponding Load Resistance:_____

Figure: Table for the Characterization measurement

Draw the characteristic curves U vs I at the given tilt angles. Find the maximum power point (MPP) of each curve. What is the corresponding load resistor for the MPP?

3.8. Task 3: Determine the optimum tilt Angle and Load Resistance

Tasks:

At the time of 12.00 using the panel holder, find the optimum tilt angle of your solar panel with respect to the sun. Measure a characteristic curve of your solar panel and derive an optimum load resistor. This is the one that produces the maximum power. Measure the solar irradiance at your measurement location.

3.9. Task 4: Measure solar generated energy at fixed tilt and load

Task:

Using the tilt and the load resistor from Chapter 3.5, measure the electrical energy generated by your solar panel over 30 minutes. Take a measurement every 5 minutes and summarize the generated energy using the table below.

Starting Time of the Measurement: _____

Tilt of the panel:_____

Load Resistance: _____

Measurement Time	Voltage	Current	Power
0 min			
5 min			
10 min			
15 min			
20 min			
25 min			
30 min			

From these measurements derive the total electrical energy generated by the solar panel.

Total Electrical Energy: _____

3.10. Dynamic Solar Panel Positioner (SPP)

To suggest possible solutions to the above questions (Chapter 3.1, 3.2), and any others that may arise in the future, we propose an experimental study based on an Arduino-controlled solar positioner system.

The Arduino project on which we've based our approach uses additive manufacturing (3D printing) to produce an initial prototype for a bi-directional (inclination and orientation) solar panel positioner. Although the approach in this project differs from the following videos, it will be helpful to watch the following videos to understand the general principle.

Link to video explanations:

Dual Axis Solar Tracker Arduino Project Using LDR & Servo Motors (Youtube: IoT Projects Ideas)

Don't forget to install the Arduino IDE to use the program (installation on computer or online).

Software: Arduino IDE

Electronics

In this project, you'll be using an INA219 measurement board to track sunlight and direct the solar panel to areas that increase their efficiency. Alternatively, LDR resistors could be used for the same purpose. The resistances of these sensors are variable and depend on the illuminance received by the sensor. However, when using the INA219 board, the voltage and current generated by the solar panel are continuously measured and two servomotors are controlled to move the solar panel in the optimum direction. An Arduino UNO R4 WiFi board (or alternatives) is used to control the servo motors. It is advisable to use a Breadboard to build and test the electronics of the project. The following figure shows the overall electronics used in this project.

You find datasheets and Tutorials related to the electronic components in the Moodle and workadventure Libraries.

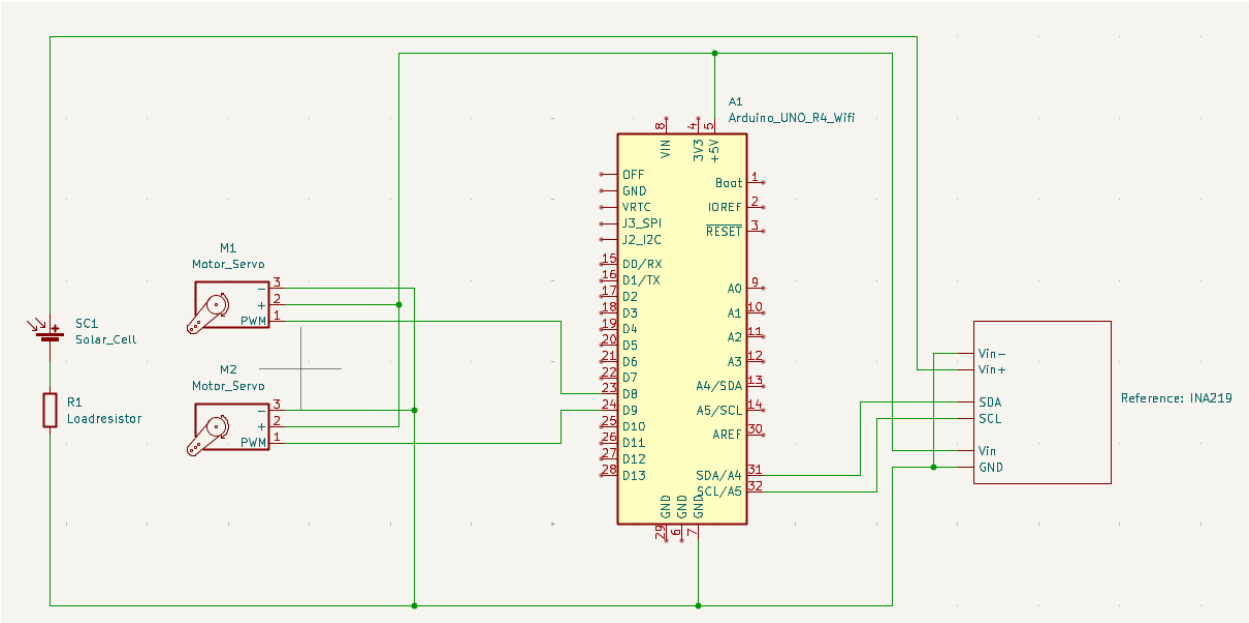


Figure X: Arduino electronics for the solar tracker

3D Models

The necessary parts for the SPP can be printed out using 3D printers available at THM and CESI. The STL files of the CAD models provided must be prepared using an appropriate Slicer Software for the 3D printer to be used (e.g. Bambu Studio for the Bambu Lab A1 mini available at THM open.A5 Makerspace). The additive manufacturing model files are available here:

ENTER STL FILE NAME OR LINK FOR 3D PRINTING

Alternatively, there is also a model file available for laser cutting from 3mm wood sheets. This can also be used but is more complex to assemble and requires additional screws which are not listed in the table below. The model files for laser cutting are available here:

LINK OR NAME OF SVG and PDF FILES FOR LASER CUTTING

The experiments were pre-tested only on the 3D-printed model.

List of Material (3D printed Model)

No.	Component	Quantity	Info
1	Arduino UNO R4 Wifi	1	Alternatives: Freenode Control Board V5
2	Servo Motor 9g, Type FS90	2	Reuse from last year.
2a	Servo Motor 9g, Type SG90	2	ALTERNATIVE: SG90 works as well, but needs adaptation of the 3D printing models
3	Solar Panel 80mm x 180mm, Vmax: 6.4V, Imax: 360mA, Pmax: 1.98W	1	Reuse from last year
4	INA219 I2C Bidirektional Current Sensor Board	1	Description
5	Load Resistor 10 Ohm, 2W	2	Load Resistor to characterize the solar panel. Value can be similar, but load must be at least 2W
6	Load Resistor 15 Ohm, 2W	2	see above
7	Load Resistor 22 Ohm, 2W	2	see above
8	Load Resistor, 47 Ohm, 2W	2	see above
9	Load Resistor, 100 Ohm, 2W	2	see above
10	Load Resistor, 270 Ohm, 2W	2	see above
11	Load Resistor, 560 Ohm, 2W	2	see above
12	Set of M2 Metal Screws (6..20mm Length)	1	Reuse from last year. Mounting the Mechanics
13	Set of 2mm Wood Screws	1	Reuse from last year. Securing the Servo Motors

14	Set of Jumper Cables m-m, f-m, f-f	1	
15	LCD Display 4x20 or 2x16 with I2C Bus for Arduino	1	OPTIONAL This, or a similar display can be used to visualize the voltage and current measurement of the solar panel instead of using the serial monitor.
16	Wooden Board as a Base 30cm x 40cm (>=10mm Thick)	1	
17	USB C Cable for Programming	1	
18	400 pin Breadboard	1	

3.11. Task 5: Build the dynamic Solar Panel Positioner

Once you've 3D printed or laser-cut your model, you'll assemble the various mechanical components of the model, including the servomotors and photovoltaic panel. Then connect the servo motors to the Arduino board.

Be aware, that you must position the servo motors correctly before you finally assemble them in the 3D printed components. You find an example code to position the servo motors in the library.

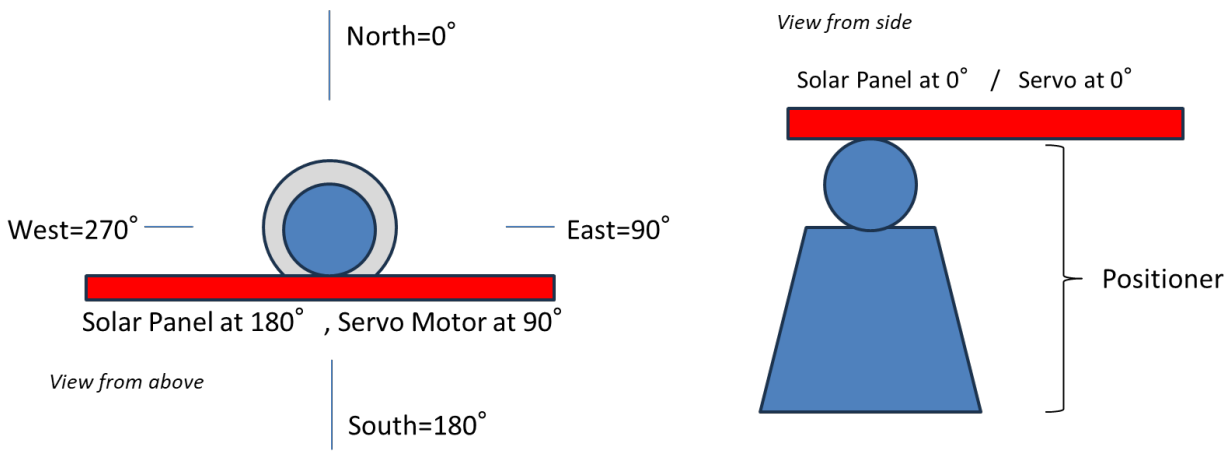
Example Sketch for Servo Motor Positioning

You will find additional examples for Arduino code from which you can start to develop the code for your prototype in the library.

Tasks:

Program the Arduino so that it positions the solar panel in a defined way.

The solar panel should be oriented towards south (180°) if the servo motor for the orientation is at 90°. The solar panel should be flat when the servo motor for the tilt is at 0°. See figure below.



Program the solar positioner in a way that it positions the solar panel at different orientations and tilts and leaves it there for 5 seconds. The positions are shown below.

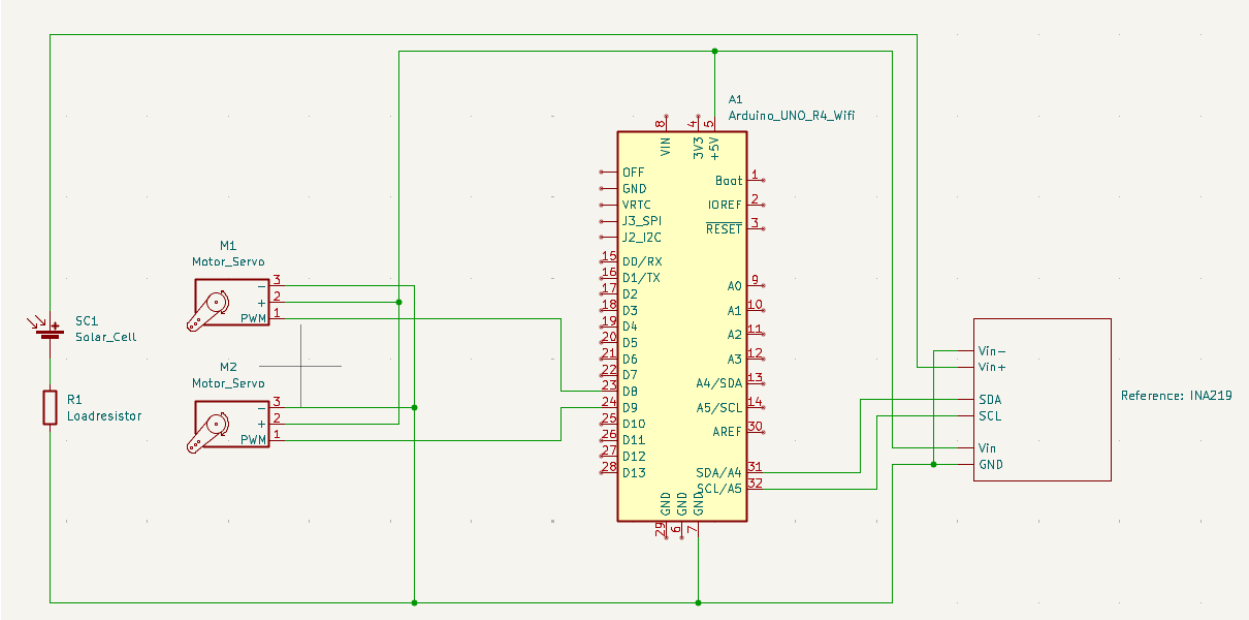
Orientation	Tilt	Duration
East (90°)	0°, 45°, 90°	5s
South (180°)	0°, 45°, 90°	5s
West (270°)	0°, 45°, 90°	5s
Optimum towards Sun	0°, 45°, 90°	5s

Once the code has been injected into the board's microcontroller, you can check that the Solar Positioner is working properly. You can modify the code to optimize your prototype.

3.12. Task 6: Measure the electrical power using the INA219 board

Now that you have completed the positioning of your solar panel, you can start to automatically measure the electrical power from the solar panel using the INA219 board.

Connect the INA219 board to your SPP in the way shown below:



You use the I2C connection of your Arduino board for this. The connection is made using the SDA and SCL pins of the INA219 board and the Arduino.

Using the example code, you can measure the voltage and the current from the solar panel using the Arduino. The results are transferred to the serial monitor of your Arduino IDE and/or to the LCD panel, if you are using one.

Example Sketch for the INA219 Board

The LCD panel can also be connected to the SDA and SCL pins in parallel to the INA219 board. The LCD must have a different address than the INA219 board (which is the case with the standard addresses).

Example Sketch for the LCD Panel

Tasks:

Bring the SPP outside and position it like described in chapter 3.8. towards north (=0°). Connect the optimum load resistor derived in chapter 3.5. Measure the voltage, current and power generated by the solar panel at the angles and tilts given in the table below, using the SPP. Record the measurements.

Starting Time of the Measurement: _____

Solar irradiance:_____

Load Resistance: _____

Orientation	Tilt	Voltage	Current	Power
140	0, 20, 30, 40, 60			
150	0, 20, 30, 40, 60			
160	0, 20, 30, 40, 60			
170	0, 20, 30, 40, 60			
180	0, 20, 30, 40, 60			
190	0, 20, 30, 40, 60			
200	0, 20, 30, 40, 60			
210	0, 20, 30, 40, 60			
220	0, 20, 30, 40, 60			

Generate a diagram showing the generated power vs the orientation and tilt.

3.13. Task 7: Automated Solar Panel Positioner

Now you should further extend your Arduino code to automatically position the solar panel to produce its maximum power related to a moving light source. So, the automated SPP should

- first find a starting position that generates the maximum power from the solar module independent from where the solar module is positioned at the start.
- Then measure voltage, current and power from the solar module continuously.
- Readjust the position of the solar module, when the light source moves away from its original position.

By realizing this system, measurements over a longer period of time can be carried out without the need for a human to position the solar module.

Tasks:

Describe the algorithm to find the initial position of the solar module which generates the maximum power on a given load resistor.

Realize the code for this algorithm and test it on the Arduino.

Describe an algorithm to detect that the light source has moved away from it original position and readjustment of the solar panel position.

Realize the algorithm in code and test it on the Arduino.

3.14. Task 8: Transfer the measured values to an MQTT broker

Now that you have realized an automated solar panel positioner with tracking function, we will have to transfer the measured values of your solar panel to a central message broker (MQTT-Broker). This allows us to collect all the power measurements from all teams in one location and be able to visualize the local and total generated electrical power.

To achieve this, you have to work on the following tasks:

- Realize a wifi connection of your Arduino board to the network of your university. You find an example sketch for the Wifi connection in the library.
- Then you have to set up an MQTT communication to the MQTT broker based on this wifi connection.
- Then you have to send the actual measurement data (current, voltage, power) to the MQTT Broker using the right topics described in chapter 2.3.3.
- Using the MQTT Explorer you can verify that the data arrived at the MQTT broker.

We will provide a software to the teams with which you can collect the measurement data from all teams and visualize it.

3.15. Node Red Dashboard

Finally, you should monitor the total generated power of all connected solar trackers between 12.00 and 12.30 and calculate the total generated power of this virtual power plant.

3.16. Conclusions and Analysis of the Automatic Solar Positioning System and Virtual Power Plant

Following the experiments conducted by all collaborators, we expect critical analysis and technical and organizational conclusions regarding the progress of this project.

Here are some suggested areas for your analysis:

- **Conclude on the feasibility of solar tracking systems (one- or two-dimensional) for solar panels and photovoltaic field installations.**
- **Propose ideas for optimizing the Energy Produced / Energy Consumed ratio in the case of a solar tracker (control system, materials).**
- **How did you define your experimental protocols? Based on which criteria did you justify the choices you made?**
- **How did you ensure the repeatability and reproducibility of your measurements? What difficulties were encountered, and what improvement areas were identified?**
- **What do you think of the "Solar Panel Positioning" prototype? What difficulties did you encounter, and what improvement areas can be identified?**
- **Conclude your work with a general conclusion on this project and provide feedback on this international collaborative effort.**
- **...**

3.17. Useful Literature

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